Program : **Diploma in Civil and EnvironmentalEngineering**	
Course Code: **5359**	Course Title: **Environmental Engineering Lab**
Semester: **5**	Credits: **1.5**
Course Category: **Programme Core Course**	
Periods per week: **3 (L:0 T:0 P:3)**	Periods per semester: **45**

Course Objectives:

- To enable students to determine physical and chemical parameters of water.
- To enable students to familiarize with water and waste water treatment steps.
- To enable students to perform air quality monitoring.

Course Prerequisites:

Topic	Course code	Course name	Semester
Fundamentals of engineering chemistry and water characteristics	1004	Applied Chemistry	1
Analytical techniques in chemistry	4352	Environmental Chemistry and microbiology	4
Experiments on water and waste water characteristics	4356	Environmental Chemistry and microbiology	4

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive level
CO1	Determine Physical and chemical characteristics of water.	9	Applying
CO2	Determine chemical characteristics of water using spectrophotometer.	9	Applying
CO3	Practice water treatment in lab and visit treatment facilities	12	Applying
CO4	Perform Air quality monitoring.	6	Applying

	Open ended experiments	3	Applying
	Lab tests	6	Applying

CO-PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3			3	2		
CO2	3			3	2		
CO3	3			3	2		
CO4	3			3	2		

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	**Determine Physical and chemical characteristics of water.**		
M1.01	Study of pH meter and Determination of pH of given water sample	3	Applying
M1.02	Determination of conductivity and TDSof given water sample	3	Applying
M1.03	Determination total hardness of water by EDTA method.	3	Applying
CO2	**Determine chemical characteristics of water using spectrophotometer.**		
M2.01	Determination of sulphate content of given water sample	3	Applying
M2.04	Determination of nitrogen of given water sample using spectrophotometer.	3	Applying
M2.03	Determine hexavalent chromium in water using spectrophotometer.	3	Applying
	Lab test 1	3	
CO3	**Practice water treatment in lab and visit treatment facilities**		

M3.01	Determine the amount of soda lime required to remove hardness in water.	3	Applying
M3.02	Removal of any one chemical parameter in water through adsorption and determination of removal efficiency. (Batch study or column study).	3	Applying
M3.03	Visit any water treatment facility and prepare report.	3	Applying
M3.04	Visit any sewage treatment facility and prepare report.	3	Applying
CO4	**Perform Air quality monitoring**		
M4.01	Estimate amount of particulate matter in ambient air.	3	Applying
M4.02	Estimate the amount of CO, CO2 and O2 using Orsat apparatus	3	Applying
	Open ended experiments	3	Applying
	Lab test- II	3	

Text / Reference:

T/R	Book Title/Author
R1	Clair N Sawyer, Perry L McCarty, Gene F Parkin- Chemistry for Environmental Engineering and Science- Mc Graw Hill
R2	IS 3025- Methods of sampling and tests of water and waste water.
R3	Standard methods for the examination of water and wastewater-American Public Health Association; American Water Works Association; and Water Pollution Control Federation U.S.A
R4	Metcalf & Eddy, Wastewater engineering: treatment and reuse, Tata McGraw Hill
R5	IS 1622-Methods of sampling and microbiological examination or water
R6	IS 5182 Indian standards – Methods for measurement of air pollution
R7	CPCB, Guidelines for the Measurement of Ambient Air Pollutants

Online Resources:

Sl.No.	Website Link
1	NITTTR NCTEL (nitttrchd.ac.in)
2	Welcome to Virtual Labs - A MHRD Govt of India Initiative (vlabs.ac.in)
3	Welcome to Virtual Labs - A MHRD Govt of India Initiative (vlabs.ac.in)